Fady Fawzy Youssef

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Summary

Computer Engineering student at Cairo University with strong interests in artificial intelligence, machine learning, embedded systems, and software engineering. Ranked third in the Preparatory Engineering year with a GPA of 3.94, and ranked first in the Computer Engineering Department with a GPA of 3.88. Experienced in building end-to-end machine learning pipelines, bare-metal embedded applications, and modular software solutions. Proven ability to translate theoretical foundations into functional technical solutions through hands-on development and cross-functional teamwork.

Education

Cairo University, Faculty of Engineering Giza, Egypt
B.Sc. in Computer Engineering (Expected 2029) 2025 – 2029

- **Academic Standing:** Ranked 1st in Computer Engineering Dept. (GPA: 3.88); Ranked 3rd in Prep Year (GPA: 3.94).

Experience & Achievements

AI & Machine Learning Intern Sep 2026 – Oct 2026
Akhnaton Trading & Distribution, EVA Group Cairo, Egypt

- Applied end-to-end machine learning workflows to real-world commercial datasets using Python, Pandas, and Scikit-learn.
- Executed exploratory data analysis (EDA), data imputation, feature scaling, and categorical encoding.
- Trained, evaluated, and tuned classification models (Logistic Regression, Decision Trees, Random Forests) using GridSearchCV.
- Structured ML models into modular, reproducible pipelines designed for clean API integration and batch inference.

Power BI Trainee 2025 – Present
Digital Egypt Pioneers Initiative (DEPI) Egypt

- Developed interactive dashboards and analytical reporting workflows to transform raw tabular data into business intelligence.
- Built data models, calculated DAX metrics, and designed clear data visualizations for performance tracking.

Research Participant — 4th Place 2026
TCCD Research Day (15th Edition), Computer Department Giza, Egypt

- Achieved 4th place departmental recognition for presenting original systems-level research.
- Co-authored and presented *Adaptive Low-Rank Memory Compression Framework for Operating Systems.*
- Investigated low-overhead compression techniques to boost main-memory efficiency and lower paging overhead.

HR Team Member 2025 – Present
IEEE CUSB (Cairo University Student Branch) Giza, Egypt

- Supported logistics, member onboarding, and organizational coordination for department events and workshops.
- Handled internal communication channels and tracked operational milestones across committee members.

Electrical & Technical Intern Jun 2025 – Oct 2025
EVA Pharma Giza, Egypt

- Assisted engineering teams in industrial facilities, backup generator testing, and power distribution panels.
- Supported technical documentation, schematic reviews in AutoCAD, and inter-departmental operational reports.

Production Line Operator (Summer Trainee) Jun 2025 – Aug 2025
EVA Pharma Giza, Egypt

- Operated high-throughput pharmaceutical packaging machinery under Good Manufacturing Practice (GMP) standards.
- Collaborated with maintenance engineers and supervisors to handle minor mechanical stoppages and minimize downtime.

Projects

Customer Churn Prediction System Repository | 2026

- Built a production-ready ML classification pipeline to predict customer churn, eliminating data leakage using Scikit-learn's `ColumnTransformer`.
- Optimized Logistic Regression, Decision Trees, and Random Forests using GridSearchCV; achieved **77% validation accuracy** and **0.80 ROC-AUC** via optimal threshold tuning.

- Developed a lightweight FastAPI backend delivering single-profile and batch CSV predictions, probabilistic risk scoring, and interactive documentation.

Smart Nurse Robot Repository | 2026

- Engineered an autonomous medical assistant using pure bare-metal ARM Thumb-2 Assembly on STM32 (without HAL or vendor libraries).
- Implemented low-level hardware drivers for I2C, SPI, UART, ADC, PWM, and external interrupts from scratch.
- Interfaced vital-sign sensors (Heart Rate, SpO2, IV drop rate, pressure), RFID authentication, and ultrasonic obstacle avoidance.
- Designed a live TFT telemetry interface for patient vital display alongside wireless Bluetooth status transmission.

Flowchart Designer and Simulator Repository | 2025

- Developed a full-featured desktop flowchart application in modern C++ without high-level external UI frameworks.
- Implemented undo/redo operation history, modular block connectors, logic validation, and automated direct C++ code generation.

Restaurant Management System Repository | 2026

- Designed an event-driven C++ simulation system modeling dine-in, takeaway, and delivery order cycles.
- Implemented custom data structures and scheduling algorithms to dynamically assign orders to chefs, tables, and delivery couriers.

FPGA 2-Phase ALU Logic Design Academic Project | 2025

- Designed, verified, and synthesized an Arithmetic Logic Unit (ALU) using Verilog HDL on the Altera DE1-SoC FPGA board.
- Supported arithmetic and logical operations, BCD-to-Excess-3 conversions, and registered outputs; verified functionality in ModelSim.

Sustainable Innovative Lampshade Media | 2026

- Led a team in designing a functional lampshade using recycled PVC materials following circular economy principles.
- Achieved **3rd place** across the department and selected as a permanent showcase model by the faculty department head.

SDG 9 Research & Keynote Presentation 2025

- Led a research team exploring UN Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure).
- Delivered the keynote address on technical innovations in sustainable infrastructure, earning outstanding evaluation marks.

Certifications

Fundamentals of Machine Learning — NVIDIA / Coursera Jun 2026
Verify Credential

Data Analysis with Python — freeCodeCamp 2025
Verify Credential

Skills & Languages

- **Programming Languages:** C, C++, Python, JavaScript, HTML5, CSS3, SQL, Verilog HDL, ARM Thumb-2 Assembly
- **AI & Data Science:** Scikit-learn, Pandas, NumPy, Exploratory Data Analysis (EDA), Model Evaluation, Hyperparameter Tuning
- **Backend & Embedded:** FastAPI, REST APIs, Bare-Metal STM32, Peripherals (UART, SPI, I2C, ADC, PWM)
- **Tools & Frameworks:** Git, GitHub, VS Code, Keil μVision, ModelSim, Power BI, AutoCAD
- **Soft Skills:** Technical Leadership, Adaptability, Cross-Functional Teamwork, Public Speaking, Problem Solving
- **Languages:** Arabic (Native), English (Professional Working Proficiency)